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Re: Initial Post

by Mansour Saeed Mubarak Hamdan Al Hamdani - Tuesday, 10 February 2026, 9:50 AM

Hi Abdulla
It is a good post, well organized, and your way of linking the project management principles and the transition of Industry 4.0 to Industry 5.0 is a great one.
In my opinion, your sample of the Maersk NotPetya incident is especially successful in terms of project governance. It is a clear demonstration that the digital transformation projects can fail not due to the bad technology, but due to the absence of risk planning gaps, redundancy, and human control (Capano, 2023). Your argument of centralized digital infrastructures forming single points of failure is important, particularly when it comes to global logistics, where failure has an immediate economic and social impact.
To answer your question, a possible approach that logistics organizations may implement to achieve efficiency by Industry 4.0 and Industry 5.0 values is to view resilience as one of the project's final deliverables, and not as an auxiliary risk control measure (Ivanov, 2023). This implies that critical aspects of cybersecurity, decentralized system architecture, and scenario-based stress testing should be integrated into project planning stages, rather than IT functions only (Bello, Jummai Okikiola, et al. 2025). The other issue is the importance of human-in-the-loop decision making, where the employees are trained to use manual or semi-digital fallback processes in the event of system failure (Tran, Tuan-Anh, et al. 2022).
Lastly, cross-functional governance can be used to strengthen sustainability and human-centricity by applying digital initiatives that are not assessed based solely on cost and performance, but also on continuity, workforce, and long-term system trust (Pourkhodabakhsh & Khajeh, 2026). On the whole, your post shows the reasons why Industry 5.0 does not represent an anti-digitization movement, but a crucial change in the approach to the organization of multi-faceted projects.
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Re: Initial Post

by Mansour Saeed Mubarak Hamdan Al Hamdani - Tuesday, 10 February 2026, 9:40 AM

Hi Andreea

This is a great post, and I appreciate that you clearly differentiate between capability and responsibility in digital transformation.

I find your presentation of Industry 4.0 as an enabler and Industry 5.0 as a corrective lens particularly enticing to the regulated industries, such as the pharmaceutical industry. The argument that even the highly automated systems may fail in socially consequential manners definitely seems to resonate, as it is a challenge to the idea that efficiency and control are the automatic equivalents of safety. Just as you point out, digital interdependence may increase risk as much as it may enhance performance (Oppenheimer, 2025).

A good example of this tension is the NotPetya one. It demonstrates that the failures of the information systems are not the technical disruptions but the systemic shocks that spread throughout the supply chains, weakening the trust, the outcome of public health, and the credibility of the institutions (Krasznay, 2020). Personally, I believe that your point that the notion of cyber-resilience, recovery capacity, and even manual workarounds are to be included in the concept of system performance is especially crucial. The resilience is far too often considered as a contingency concept, instead of a design consideration (Bendiek & Schulze, 2021).

The best thing is your closing question, which is effective to whom and strong in what conditions, which is such an excellent definition of the normative change that Industry 5.0 is demanding. Industry 5.0 does not deny the concept of digital transformation but instead repackages the notion of success to curb the demands of societies and long-term system stability through the inclusion of human-centricity, sustainability, and resilience in the performance criteria (Harmash, O. et al. 2024).

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Peer Response

by Ioanna Tziouvaka - Monday, 16 February 2026, 3:38 PM

Hi Andreea,

Your use of the NotPetya cyberattack is a strong way to demonstrate how deeply interconnected manufacturing systems have become under Industry 4.0. The example clearly shows that when production relies on tightly integrated digital infrastructures, a disruption at one supplier can quickly escalate into system-wide consequences. This reflects broader research showing that digitally connected supply networks tend to magnify shocks rather than contain them, particularly when optimisation models are designed around stability rather than disruption tolerance (Ivanov, Dolgui and Sokolov, 2019).

What I find particularly interesting in your argument is the implication this has for how we evaluate machine learning (ML) in manufacturing. In many industrial settings, model performance is assessed through metrics such as accuracy, recall, or downtime reduction. However, these indicators do not necessarily tell us how models behave when data pipelines are interrupted or corrupted, which is precisely what happens during cyber incidents (Amodei et al., 2016). A scheduling or quality-control model may perform extremely well under normal operating conditions, yet prove fragile when faced with distrib[illegible]d by sensor failure or malicious interference. Strengthening ML pipelines through adversarial testing and unce[illegible]ld therefore enhance resilience at a system level (Goodfellow, Shlens and Szegedy, 2014).

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